RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

B.Sc. (General) Degree in Applied Sciences

First Year Semester I Examination– May/ June 2016

FDN 1204 – Basic Mathematics

Time: 02hours

Answer four (04) questions, including the question number one (01).

1. Given that, $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}$ $B = \begin{pmatrix} 0 & -2 & -3 \\ 0 & 0 & -1 \\ -2 & 0 & 0 \end{pmatrix}$ $C = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$ $D = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

   (a) Find A+B and what is the uniqueness of the resulted matrix?

   (b) Find, (i) A^T (ii) D^{-1} (iii) $A+B^T$ (iv) AC
   (v) CA (vi) $D^{-1}D$ (vii) DD^{-1}

   (c) Explain the property of D and D^{-1} matrices.

(25 Marks)

2. (a) Simplify and express in the form of $(a+bi)$, $\frac{3i^8-i^{11}}{2i-\sqrt{3}}$

   (b) Find the real numbers x and y such that $i^3 + 6y + 3ix - iy + 5x = -1 + 3i$

   (c) (i) What is the complex conjugate of $e^{i\theta} = \cos\theta + i\sin\theta$?

   Given that, $e^{i\theta} = \cos\theta + i\sin\theta$
   $e^{-i\theta} = \cos\theta - i\sin\theta$

   Show that, (ii) $\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$ (iii) $\sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$
   (iv) $\sin^2\theta + \cos^2\theta = 1$ (v) $\cos 2\theta + i\sin 2\theta = e^{2i\theta}$

   (d) Given that $\log_5(x^3 + x^2 - 9) = 3\log_5 x$, find the value x.

(25 Marks)

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3. (a) In a combustion reaction, the mass (in gram) of fuel decreased according to the formula $M = 4 - 0.05t^2$ (t in hours).
   (i) Find the average rate of reaction in the interval from $t = 2$ to $t = 3$.
   (ii) Find the rate of reaction at $t = 4$.
   (iii) State whether the curve of combustion reaction is parabola or a straight line.
   (iv) Find the turning point of this curve and determine whether it is a maximum, minimum or a point of inflexion.

   (b) Given that Arrhenius equation as $k = Ae^{-\frac{E_a}{RT}}$ where, k is rate of the reaction, T is the temperature, R is the Universal gas constant, A is pre-exponential factor and E_a is activation energy.
   (i) Express the equation in linear form using logarithms.
   (ii) If $\frac{1}{T}$ is the independent variable and $\ln k$ is the dependant variable convert the Arrhenius equation into the form of $y = mx + c$.
   (iii) Write the values of gradient (m) and intercept (c).
   (iv) Write down the 1st partial derivative of Arrhenius equation with respect to T.

(25 marks)

4. (a) Solve the equation $x^4 - 7x^2 + 12 = 0$

   (b) Find the area of the region bound by the x-axis $\sqrt{3}$ and 2 of the curve $y = x^4 - 7x^2 + 12$, explain the sign of the area obtained.

   (c) In chemical thermodynamics the G = H – TS where, G = Gibbs free energy, H = enthalpy, T = temperature and S = entropy are state variables. The infinitesimal change can be denoted as (G+∂G), (H+∂H), (S+∂S) and (T+∂T).
   (i) Show that $\partial G = \partial H - T\partial S - S\partial T$.
   (ii) Find all partial derivatives of G = H – TS.
   (iii) Write the total differential of the function G = H – TS.

(25 Marks)

5. (a) Factorise the equation $3x^3 + 4x^2 - 4x$

   (b) Integrate $\int \frac{x^2+4}{3x^3+4x^2-4x} dx$

   (c) Determine $\frac{dy}{dx}$ from 1st principle of the function $y = 3x^2 + 5$.

(25 Marks)

① $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ $\quad B = \begin{bmatrix} 0 & -2 & -3 \\ 0 & 0 & -1 \\ -2 & 0 & 0 \end{bmatrix}$ $\quad C = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$ $\quad D = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

a) $A + B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -2 & -3 \\ 0 & 0 & -1 \\ -2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

ඒකක න්‍යාසයකි (I_n)

b) i) $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix}$

ii) $D^{-1} = \dfrac{1}{4-6} \begin{bmatrix} 4 & -2 \\ -3 & \cancel{+}1 \end{bmatrix}$

$= \dfrac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

$= \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$

iii) $B^T = \begin{bmatrix} 0 & 0 & -2 \\ -2 & 0 & 0 \\ -3 & -1 & 0 \end{bmatrix}$

$A + B^T = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & -2 \\ -2 & 0 & 0 \\ -3 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix}$

(iv) $AC = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}_{3\times 3} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}_{3\times 1}$

$= \begin{bmatrix} 1\times 3 + 2\times 1 + 3\times 0 \\ 0\times 3 + 1\times 1 + 1\times 0 \\ 2\times 3 + 0\times 1 + 1\times 0 \end{bmatrix}_{3\times 1} = \begin{bmatrix} 3+2+0 \\ 0+1+0 \\ 6+0+0 \end{bmatrix}$

$= \begin{bmatrix} 5 \\ 1 \\ 6 \end{bmatrix}_{3\times 1}$

(v) $CV = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}_{3\times1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}_{3\times3}$

CA

CA ta gunithyak netha.

$$= \begin{bmatrix} \cancel{3\times1} & \cancel{3\times2} & \cancel{3\times3} \\ 1\times1 & 1\times2 & 1\times3 \\ \cancel{0\times1} & \cancel{0\times2} & \cancel{0\times3} \end{bmatrix}_{3\times3} = \begin{bmatrix} \cancel{3} & \cancel{6} & \cancel{9} \\ 1 & 2 & 3 \\ \cancel{0} & \cancel{0} & \cancel{0} \end{bmatrix}_{3\times3}$$

(v) $D^{-1}D = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & \frac{-1}{2} \end{bmatrix}_{2\times2} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times2}$

$$= \begin{bmatrix} -2\times1 + 1\times3 & -2\times2 + 1\times4 \\ \frac{3}{2}\times1 + \frac{-1}{2}\times3 & \frac{3}{2}\times2 + \frac{-1}{2}\times4 \end{bmatrix} = \begin{bmatrix} -2+3 & -4+4 \\ \frac{3}{2}-\frac{3}{2} & 3-2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times2}$$ ඒකක න්‍යාසයකි $(I_n)_{2\times2}$

(vi) $DD^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times2} \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}_{2\times2}$

$$= \begin{bmatrix} 1\times-2 + 2\times\frac{3}{2} & 1\times1 + 2\times-\frac{1}{2} \\ 3\times-2 + 4\times\frac{3}{2} & 3\times1 + 4\times-\frac{1}{2} \end{bmatrix} = \begin{bmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times2}$$ ඒකක න්‍යාසයකි $(I_n)_{2\times2}$

c) $= D^{-1}D = I_{n\,2\times2}$ $\quad DD^{-1} = I_{n\,2\times2}$

$\therefore D^{-1}D = DD^{-1} = I_{n\,2\times2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(2) $(a+bi)$ ආකාරය $\dfrac{3i^8 - i^{11}}{2i - \sqrt{3}}$ [illegible]

$\sqrt{-1} = i$

$-1 = i^2$

$$\frac{i^8(3 - i^3)}{2i - \sqrt{3}} = \frac{(i^2)^4\,[3 - i(i^2)]}{2i - \sqrt{3}}$$

$$= \frac{(-1)^4\,[3 - i(-1)]}{2i - \sqrt{3}} = \frac{1\,(3+i)}{2i - \sqrt{3}}$$

$$= \frac{3+i}{2i-\sqrt{3}} \times \frac{2i+\sqrt{3}}{2i+\sqrt{3}} = \frac{6i + 3\sqrt{3} + 2i^2 + \sqrt{3}i}{(2i)^2 - (\sqrt{3})^2} = \frac{6i + 3\sqrt{3} - 2 + \sqrt{3}i}{4(-1) - 3}$$

$$= \frac{(\sqrt{3}+6)i + (3\sqrt{3} - 2)}{-7} = \frac{\sqrt{3}+6}{7}\,i + \frac{3\sqrt{3}-2}{7}$$

$$\uparrow \qquad\qquad \uparrow$$

$$b\,i \quad + \quad a$$

b) $i^3 + 6y + 3ix - iy + 5x = -1 + 3i$

$i(i^2) + 6y + 5x + (3x - y)i = -1 + 3i$

$-i + 6y + 5x + (3x - y)i = -1 + 3i$

$6y + 5x + (3x - y - 1)i = -1 + 3i$

[illegible]

$6y + 5x = -1$

$6(3x - 4) + 5x = -1$

$18x - 24 + 5x = -1$

$23x = 23$

$\underline{\underline{x = 1}}$

[illegible]

$3x - y - 1 = 3$

$3x - y = 4$

$3x - 4 = y$

ආදේශයෙන්

$3(1) - 4 = y$

$\underline{\underline{-1 = y}}$

ආදේශයෙන්

c) i) $e^{i\theta} = \cos\theta + i\sin\theta$

complex comjugate $= i\sin\theta$

ii) $\cos\theta + i\sin\theta = e^{i\theta} \longrightarrow ①$

$\cos\theta - i\sin\theta = e^{-i\theta} \longrightarrow ②$

①+②

ii) $2\cos\theta = e^{i\theta} + e^{-i\theta}$

$\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2} \longrightarrow ③$

①−②

iii) $2i\sin\theta = e^{i\theta} - e^{-i\theta}$

$\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i} \longrightarrow ④$

iv) $④^2 + ③^2$

$$\sin^2\theta + \cos^2\theta = \left(\frac{e^{i\theta} - e^{-i\theta}}{2i}\right)^2 + \left(\frac{e^{i\theta} + e^{-i\theta}}{2}\right)^2$$

$$= \frac{e^{2i\theta} - 2e^{i\theta}e^{-i\theta} + e^{+2i\theta}}{-4} + \frac{e^{2i\theta} + 2e^{i\theta}e^{-i\theta} + e^{+2i\theta}}{4}$$

$\boxed{4\,i^2 \underset{\to -1}{=4\times-1}}$

මේ -4 වෙනවා

$$= \frac{-e^{2i\theta} + 2e^{i\theta}e^{-i\theta} - e^{+2i\theta} + e^{2i\theta} + 2e^{i\theta}e^{-i\theta} + e^{+2i\theta}}{4}$$

$$= \frac{4e^{i\theta}e^{-i\theta}}{4} = e^{i\theta}e^{-i\theta} = e^{(i\theta - i\theta)} = e^0 = 1$$

$\underline{\underline{\sin^2\theta + \cos^2\theta = 1}}$ ~~$e^{i\theta}e^{-i\theta}$~~

v) $③^2 - ④^2$

$$\underbrace{\cos^2\theta - \sin^2\theta}_{\cos 2\theta} = \left(\frac{e^{i\theta} + e^{-i\theta}}{2}\right)^2 - \left(\frac{e^{i\theta} - e^{-i\theta}}{2i}\right)^2$$

$$= \frac{(e^{i\theta} + e^{-i\theta})^2}{4} - \frac{(e^{i\theta} - e^{-i\theta})^2}{4i^2} \longrightarrow 4\times-1 = -4$$

$$= \frac{(e^{i\theta} + e^{-i\theta})^2}{4} + \frac{(e^{i\theta} - e^{-i\theta})^2}{4}$$

$$\cos 2\theta = \frac{e^{2i\theta} + 2e^{i\theta}e^{-i\theta} + e^{2i\theta} + e^{2i\theta} - 2e^{i\theta}e^{-i\theta} + e^{2i\theta}}{4}$$

$$= \frac{4e^{2i\theta}}{4}$$

$$\underline{\underline{\cos 2\theta = e^{2i\theta}}}$$

d)
$$\log_5 (x^3 + x^2 - 9) = 3 \log_5 x$$
$$\log_5 (x^3 + x^2 - 9) = \log_5 x^3$$
$$x^3 + x^2 - 9 = x^3$$
$$x^2 - 9 = 0$$
$$x^2 - 3^2 = 0$$
$$(x-3)(x+3) = 0$$

$$x + 3 = 0$$
$$\underline{\underline{x = -3}}$$

හෝ

$$x - 3 = 0$$
$$\underline{\underline{x = 3}}$$

5) a)
$$3x^3 + 4x^2 - 4x$$
$$x(3x^2 + 4x - 4)$$
$$x(3x^2 + 6x - 2x - 4)$$
$$x[3x(x+2) - 2(x+2)]$$
$$\underline{\underline{x(x+2)(3x-2)}}$$

b)
$$I = \int \frac{x^2 + 4}{3x^3 + 4x^2 - 4x} \, dx$$

$$= \int \frac{x^2 + 4}{x(x+2)(3x-2)} \, dx$$

$$\frac{x^2 + 4}{x(x+2)(3x-2)} = \frac{A}{x} + \frac{B}{(x+2)} + \frac{C}{(3x-2)}$$

භින්න භාගවලට වෙන් කරන්න.

$$x^2 + 4 = A(x+2)(3x-2) + Bx(3x-2) + Cx(x+2)$$

$x = 0$ විට

$$0 + 4 = A(2)(-2)$$
$$\underline{-1 = A}$$

$x = -2$ විට

$$(-2)^2 + 4 = B(-2)(3\times -2 - 2)$$
$$8 = B \times -2 \times -8$$
$$\underline{\frac{1}{2} = B}$$

$\underline{x^2 \text{ සංගුණක}}$,

$$1 = 3A + 3B + C$$
$$1 = 3 \times -1 + 3 \times \frac{1}{2} + C$$
$$4 = \frac{3}{2} + C \qquad \frac{5}{2} = C$$

$$\frac{x^2}{x(x-2)(3x-2)} = \frac{-1}{x} + \frac{1}{2(x+2)} + \frac{5}{2(3x-2)}$$

$$\int \frac{x^2}{3x^3+4x^2-4x}\,dx = -\int \frac{1}{x}\,dx + \frac{1}{2}\int \frac{1}{(x+2)}\,dx + \frac{5}{2}\times\frac{1}{3}\int \frac{1\times 3}{3x-2}\,dx$$

$$= -\ln|x| + \frac{1}{2}\ln|x+2| + \frac{5}{6}\ln|3x-2| + K$$

K - අභිමත නියතය

c) $y = 3x^2 + 5$

$$\frac{dy}{dx} = 3(2x) + 0$$

$$= \underline{\underline{6x}}$$

ප්‍රථම මූලධර්ම භාවිතයෙන් හඳුනාගමු,

$$y = 3x^2 + 5 \longrightarrow (1)$$

Δx, Δy යනු ඉතා කුඩා ~~....~~ අගයක් යැයි ගමු.

$$y + \Delta y = 3(x + \Delta x)^2 + 5 \longrightarrow (2)$$

(2) − (1)

$$\Delta y = 3(x + \Delta x)^2 - 3x^2$$

$$\Delta y = 3(x^2 + 2x\Delta x + \Delta x^2 - x^2)$$

$$\Delta y = 3(2x\Delta x + \Delta x^2)$$

$$\frac{\Delta y}{\Delta x} = 3(2x + \Delta x)$$

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = 6\lim_{\Delta x \to 0} x + 3\lim_{\Delta x \to 0} \Delta x$$

$$\frac{dy}{dx} = 6x + 3\times 0$$

$$\frac{dy}{dx} = \underline{\underline{6x}}$$

(3) a) $M = 4 - 0.05t^2$

$t = 2$

$M = 4 - 0.05(2)^2$

$= 4 - 0.2$

$= \underline{\underline{3.8}}$

$t = 3$

$M = 4 - 0.05(3)^2$

$= 4 - 0.45$

$= \underline{\underline{3.55}}$

$t = 4$

$M = 4 - 0.05(4)^2$

$= 4 - 0.8$

$= \underline{\underline{3.2}}$

$t = 1$ විට

$M = 4 - 0.05$ ~~380 : 355~~

$= 3.95$

$t = 2$ 395 : 380

$t = 3$ 395 : 355

$t = 4$ 395 : 320

③ a) $M = 4 - 0.05t^2$

$$\frac{dM}{dt} = 0 - 0.05 \times 2t$$

$$\frac{dM}{dt} = -0.1t$$

i) $t = 2$ විට

$$\frac{dM}{dt} = -0.1 \times 2$$

$$= \underline{\underline{-0.2}}$$

$t = 3$ විට

$$\frac{dM}{dt} = -0.1 \times 3$$

$$= \underline{\underline{-0.3}}$$

මේක තෝරන්න බෑනෙ අහන්නෙ ?

ii) $t = 4$ විට

$$\frac{dM}{dt} = -0.1 \times 4$$

$$= \underline{\underline{-0.4}}$$

මේකත් මට sure නෑ .?

iii) මරුන්ට ප්‍රතිකාරයක් නිසා අඩුවෙලායකි . ✓

iv) $\frac{dM}{dt} = 0$ විට හැරුම් ලක්ෂ වෙයි .

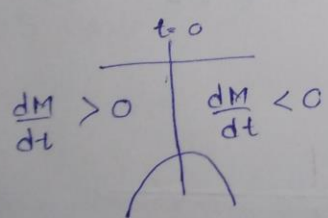
$$-0.1t = 0$$

$$t = 0$$

$t = 0$ විට $M = 4 - 0.05 \times 0$

$= 4$

හැරුම් ලක්ෂය = $(0, 4)$ ✓



∴ ~~උපරිම~~ හැරුම් ලක්ෂය උපරිමයකි . ✓

b) $\dot{K} = \dot{A}e^{\left(\frac{-\hat{E}a}{RT}\right)}$

i) $\ln K = \ln Ae\left(\frac{-Ea}{RT}\right)$

ii) $\ln K = \ln \frac{Ae(-Ea)}{R} + \ln \frac{1}{T}$

$y = c + mx$

iii) $m = 1$

$c = \ln \frac{Ae(-Ea)}{R}$

iv) ?

(a) $x^4 - 7x^2 + 12 = 0$ $\qquad x^2 = t$ යැයි ගමු.

$t^2 - 7t + 12 = 0$

~~$t^2 - 6t - t + 12 = 0$~~

~~$t(t-6) - t$~~

$t^2 - 4t - 3t + 12 = 0$

$t(t-4) - 3(t-4) = 0$

$(t-3)(t-4) = 0$

$t - 3 = 0$ හෝ $t - 4 = 0$

$t = 3$ $\qquad t = 4$

$x^2 = 3$ $\qquad x^2 = 4$

$x = \pm\sqrt{3}$ $\qquad x = \pm 2$

$x = -\sqrt{3}, +\sqrt{3}, -2, 2$

$y = x^4 - 7x^2 + 12$

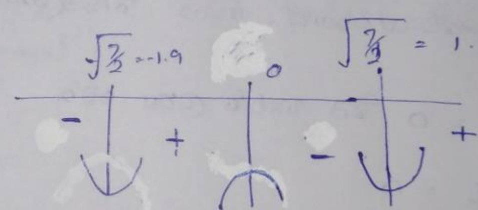
$y = 0$ විට $x = -\sqrt{3}, +\underset{1.73}{\sqrt{3}}, -2, 2$ $\qquad (-\sqrt{3}, 0)\ (\sqrt{3}, 0)\ (-2, 0)\ (2, 0)$

$x = 0$ විට $y = 12$

$\frac{dy}{dx} = 4x^3 - 14x$

$\frac{dy}{dx} = 0$ විට ස්ථාවර ලක්ෂ පවතී

$0 = x(4x^2 - 14)$



$x = 0$ $\qquad 4x^2 - 14 = 0$

$y = 12$ $\qquad x = \pm\sqrt{\frac{7}{2}}$

$y = \frac{49}{4} - 7 \times \frac{7}{2} + 12$

$= \frac{-49}{4} + 12 = \frac{-1}{4} = -0.25$

හැරුම් ලක්ෂ $\quad (0, 12)$, $\left(\sqrt{7/2}, -\frac{1}{4}\right)$, $\left(-\sqrt{7/2}, -\frac{1}{4}\right)$

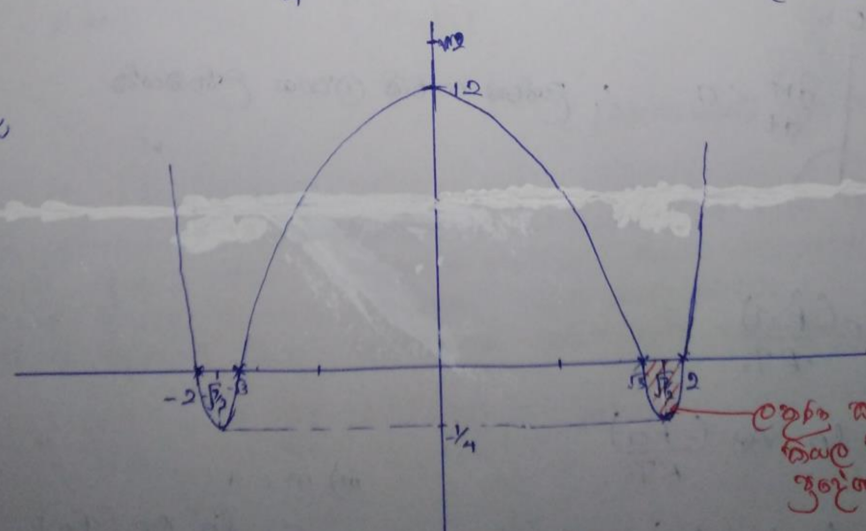
අන්ත හැසිරීම

$x \to \infty$

$y \to \infty$

$x \to -\infty$

$y \to +\infty$



ලකුණු කරන්න කියල තියෙන ප්‍රදේශය

c) $G = H - TS \longrightarrow ①$

$G + \Delta G = H + \Delta H - (T + \Delta T)(S + \Delta S) \longrightarrow ②$

ΔT, ΔS, ΔG, ΔH
ඉතා කුඩා + සංඛ්‍යා වේ.

② − ①

$\Delta G = \Delta H - (T + \Delta T)(S + \Delta S) + TS$

$\Delta G = \Delta H - TS - T\Delta S - S\Delta T - \Delta T\Delta S + TS$

$\Delta G = \Delta H - T\Delta S - S\Delta T - \Delta T \cdot \Delta S$

"0 ලෙස ගන්නවා

ΔT, ΔS ඉතා කුඩා + සංඛ්‍යා නිසා
ΔT.ΔS ඊටත් වඩා
කුඩා + සංඛ්‍යාවක් නිසා
නොසලකා හැරිය හැක.

$\therefore \Delta G = \Delta H - T\Delta S - S\Delta T$

ii)

$G = H - TS$

$[G] = kJ\,mol^{-1} = [J\,mol^{-1}]$ $[H] = kJ\,mol^{-1} = [J\,mol^{-1}]$ $[T] = [K] / [^{\circ}C]$

$[S] = [J\,mol^{-1}\,{}^{\circ}C^{-1}] / [J\,mol^{-1}\,K^{-1}]$

R.H.S,

$[G] = [J\,mol^{-1}]$

L.H.S

$[H] - [T][S]$

$[J\,mol^{-1}] - [^{\circ}C] \times [J\,mol^{-1}\,{}^{\circ}C^{-1}]$

$[J\,mol^{-1}] - [J\,mol^{-1}]$

$[J\,mol^{-1}]$

R.H.S = R.H.S

මේක sure නෑ.
මම අර්ථ ගන්වන්නේ
කියන්නෙ මොකක්ද
කියල දන්නෙ නෑ.
ඒකයි ඒකක සමාන කරල
(සමානතාවය) තහවුරු කළේ.

~~III)~~

iii) $G = H - TS$

මේක අවකලනය කරන්න කියලා කියන්නෙ. ඒත් ඒ මොන විචල්‍යයක්ද කියලා කියල නැති නිසා අවකලනය කරන්නෙ. ඔයාට තේරෙනවනම් මොකක් හරි විචල්‍යයක් කියල තියෙනවා ඒකෙන් අවකලනය කරන්න. මම x විචල්‍යයක් වගේ අවකලනය කරන හැටි පෙන්නනම්. ඒ විදිහට කරන්න.

$$\frac{dG}{dx} = \frac{dH}{dx} - T\frac{dS}{dx} - S\frac{dT}{dx}$$ මේ විදිහට.